GROUP 1 : Ferdia Comerford, Hani Ali, Zhuohan Dai

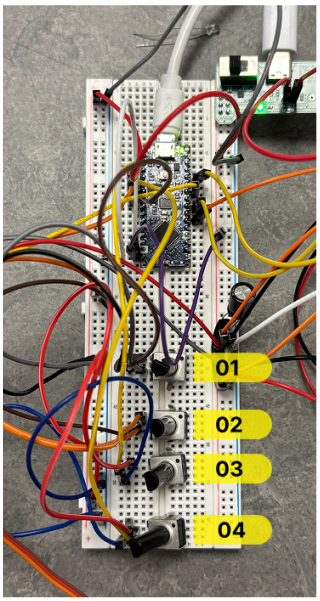
GITHUB LINK - **ZhuohanDai/Robot-Arm-and-Machine-Learning-Group-1**

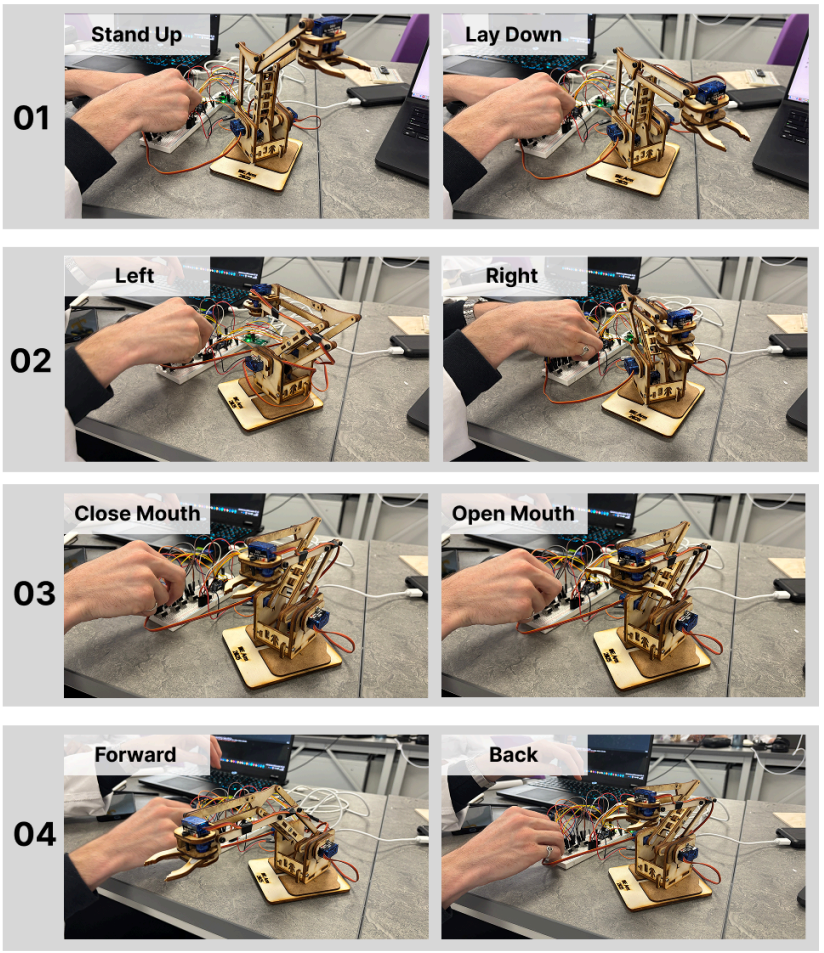
A1 - BUILT ROBOT ARM AND SYSTEM UNIT TESTS CHECK-IN

D1

DIGITAL CONTROL

1. Single unit test (Arduino sketch) for validating actuator output





2. Debugging action list / decision tree in the test documentation

1. Check Wiring — Ensure all wires are securely connected to the correct pins.

   Expected observation: All connections are secure, with no loose or misplaced wires.

2. [BLINK CODE] Upload Blink Test — Upload BLINK program to verify the Arduino is linked correctly to Processing and Wekinator. If on board LED light responds to mouse movement

the connection is correct. Check Wekinator input (mouse movement) is changing the classification in processing.

Expected observation: The LED blinks at regular intervals.

3. Test Potentiometer — Upload Potentiometer code to Ardunino. Rotate each potentiometer and observe the corresponding input readings or servo response.

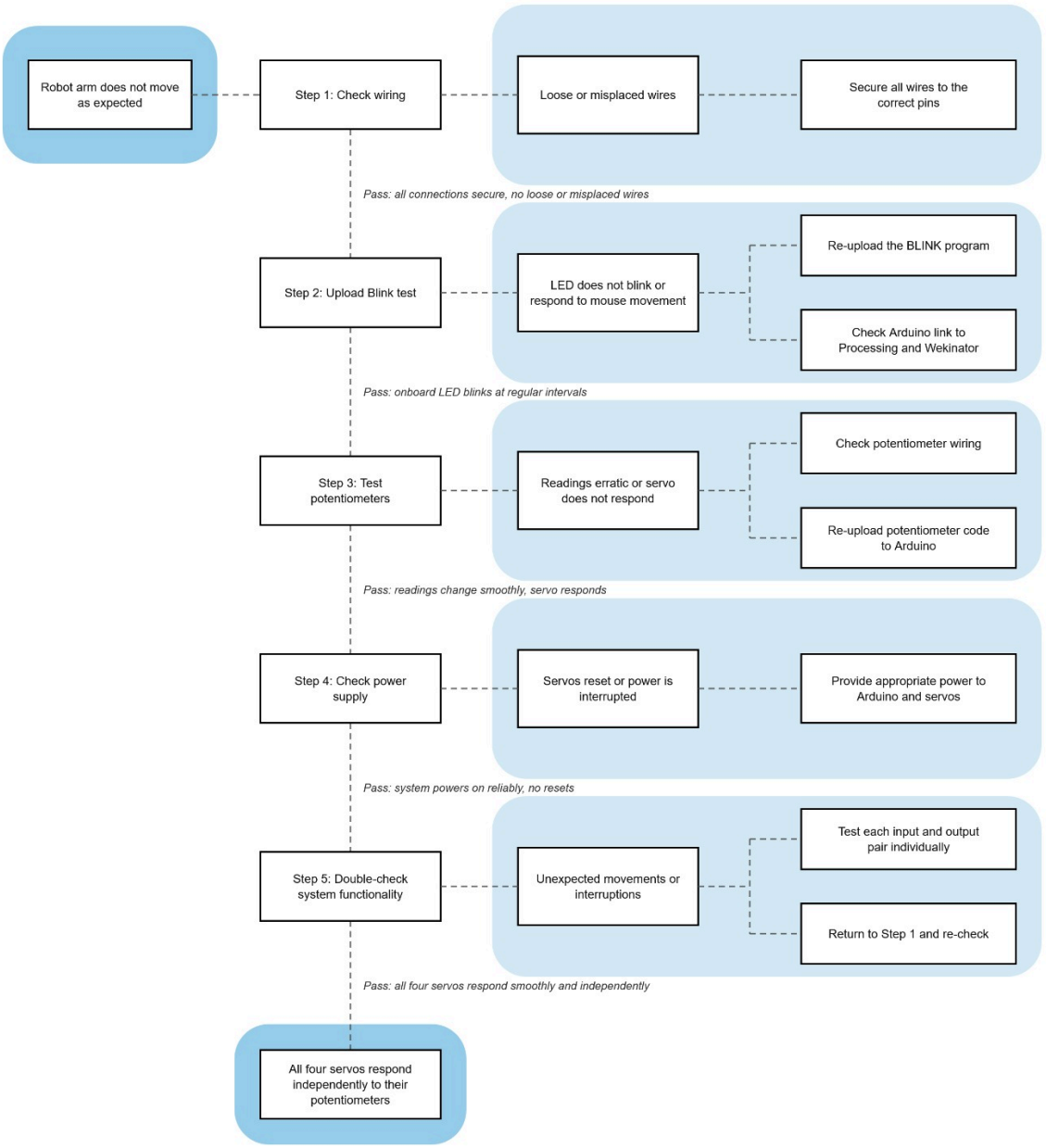
   Expected observation: The readings change smoothly as the potentiometer rotates, and the corresponding servo responds accordingly.

4. Check Power Supply — Verify that the Arduino and servos receive the appropriate power supply.

   Expected observation: The system powers on reliably, and the servos operate without unexpected resets or power interruptions.

5. Double-check System Functionality — Test all components together and check that each input produces the intended output.

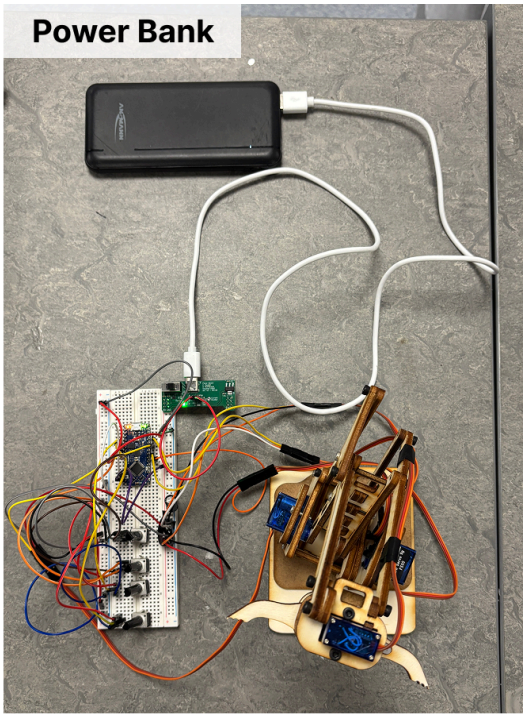
   Expected observation: All four servos respond smoothly and independently to their corresponding potentiometers, with no unexpected movements or interruptions.

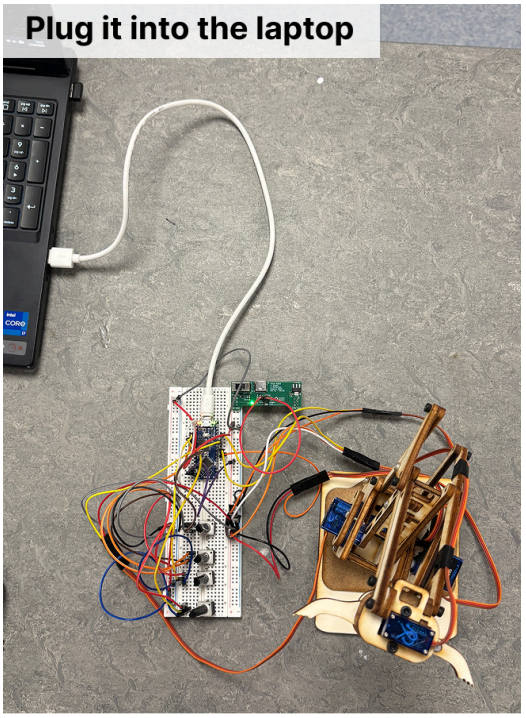

Robot arm does not move as expected
Step 1: Check wiring
Loose or misplaced wires
Secure all wires to the correct pins
Pass: all connections secure, no loose or misplaced wires
Step 2: Upload Blink test
LED does not blink or respond to mouse movement
Re-upload the BLINK program
Check Arduino link to Processing and Wekinator
Pass: onboard LED blinks at regular intervals
Step 3: Test potentiometers
Readings erratic or servo does not respond
Check potentiometer wiring
Re-upload potentiometer code to Arduino
Pass: readings change smoothly, servo responds
Step 4: Check power supply
Servos reset or power is interrupted
Provide appropriate power to Arduino and servos
Pass: system powers on reliably, no resets
Step 5: Double-check system functionality
Unexpected movements or interruptions
Test each input and output pair individually
Return to Step 1 and re-check
Pass: all four servos respond smoothly and independently
All four servos respond independently to their potentiometers

3. Manual Control of Servo Using Potentiometers

<u>Eelectronics Design</u>

- Circuit powered by power bank
- Arduino powered by laptop



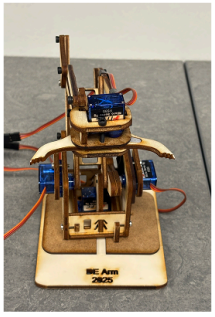
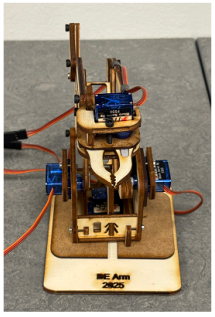


D2

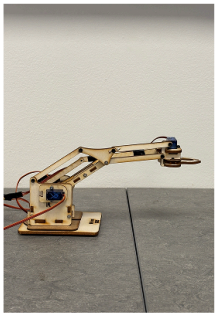
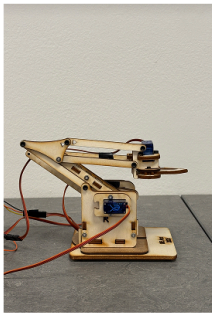
Mechanism Functioning

- Independent actuation of all 4 servos over their full range of motion

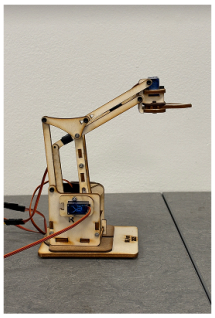
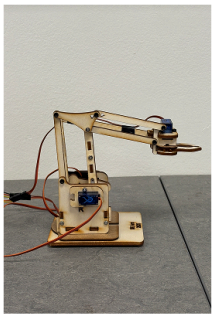
Mouth Open/ Close

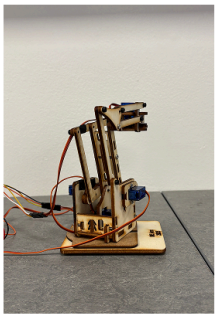
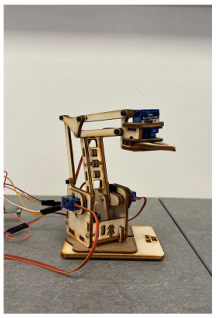
Forward/ Back

Stand Up/ Lay Down

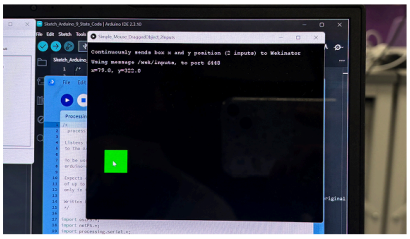
Rotate Left/ Right

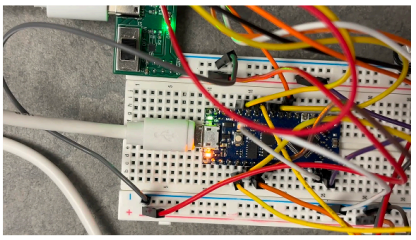



D3

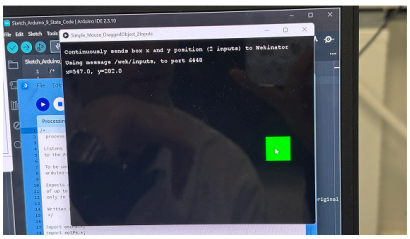
Computer-to-Arduino Output Validation (Blink Test)

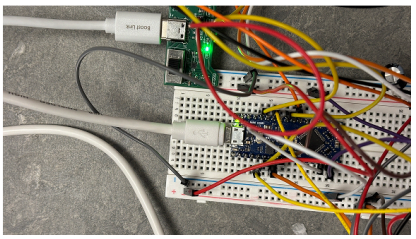
Blink quick





Blink slow





Debugging action list

1. Check Wiring — Ensure the Arduino is connected to the computer using a USB cable and that the onboard LED is working.

   Expected observation: The Arduino powers on, and the onboard LED responds when the program runs.

2. [BLINK CODE] Upload Arduino Test — Upload the blink code to the Arduino and check that the onboard LED flashes. The LED should blink using the default timing if no serial message is received.

   Expected observation: The onboard LED flashes on and off regularly.

3. [PROCESSING] Check Serial Port — Run Processing and check that `COM` matches the Arduino's actual port.

   Expected observation: Processing connects to the Arduino without serial port errors.

4. [WEKINATOR] Check Input and Classification — Start Wekinator and move the mouse or change the input. Check the Processing console to confirm that classification values are being received.

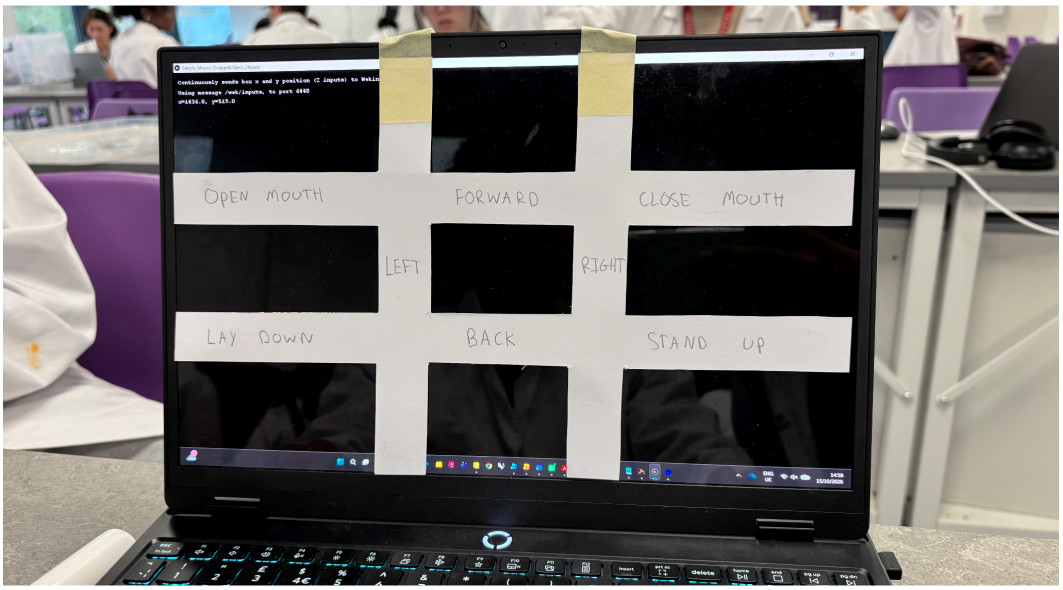
   Expected observation: The classification values change according to the input and appear in the Processing console.

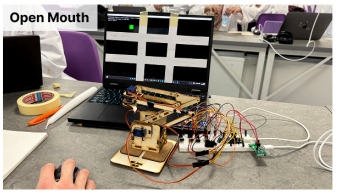
5. [BLINK PATTERN] Test All Three Classes — Test classes 1 and 2. Each class should produce a different blink pattern.

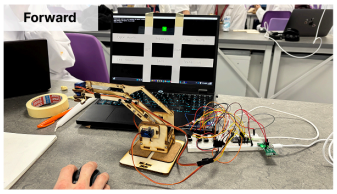
   Expected observation: The LED displays a distinct blink pattern for each class. If another value is received, the LED uses the default pattern of 1000 ms on and off.

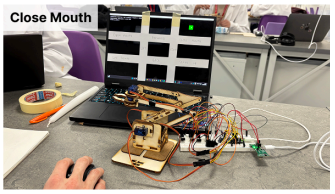
D4 - ARDUINO OUTPUT CONTROLLED BY WEKINATO

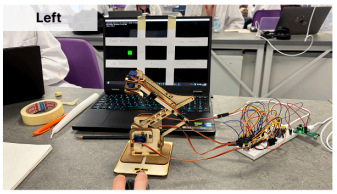
D4

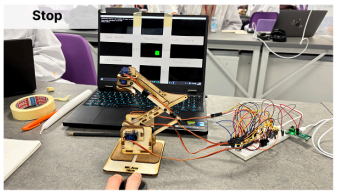


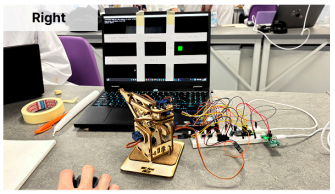


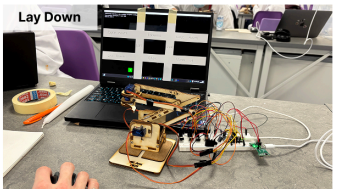


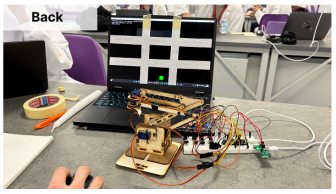


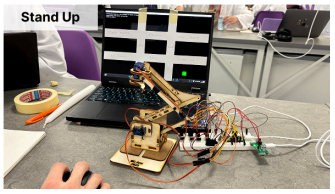












Debugging action list

1. Check Wiring — Ensure all four servos are connected to the correct Arduino pins: servo 1 to pin 9, servo 2 to pin 7, servo 3 to pin 5 and servo 4 to pin 3. Check that the power supply and ground connections are secure.

   Expected observation: All connections are secure, and the servos receive sufficient power without unexpected resets or interruptions.

2. [PROCESSING] Check Serial Connection — Run Processing and ensure that `COM` matches the Arduino's actual serial port.

   Expected observation: Processing connects to the Arduino successfully, with no serial port errors.

3. [WEKINATOR] Train and Test Nine Classifiers — Set up Wekinator with nine classes representing the intended robot arm movements. Train each class using the appropriate mouse movements or inputs, then test whether Wekinator can distinguish between them correctly.

   Expected observation: Wekinator recognises the nine classes and produces the correct classification number for each intended input.

4. [SERVO MOVEMENTS] Test All Four Servos — Test classes 1–8 individually. Classes 1 and 2 control servo 1, classes 3 and 4 control servo 2, classes 5 and 6 control servo 3, and classes 7 and 8 control servo 4. Check that each servo opens and closes as intended.

   Expected observation: Each servo responds to its assigned classes and moves smoothly between its programmed positions without unintended movements.

5. [CLASS 9] Test the No-Movement State — Send class 9 from Wekinator and check that the Arduino does not issue any new servo movement commands.

   Expected observation: All four servos remain at their existing positions when class 9 is detected.

6. Double-Check System Functionality — Run Wekinator, Processing and the Arduino together. Test all nine classifications repeatedly and check that Wekinator correctly distinguishes between inputs and that each class produces the intended response.

   Expected observation: The complete system reliably distinguishes between the nine classes, all four servos respond smoothly and independently to their assigned commands, and class 9 produces no new movement commands.